Integrated pipeline and fixed test set

Team: Marc Justin Alberto, Jose III Aquino, Jeremy Garin. Phase 3 hand-in: 29 October 2026, 11:30 PM; class check: 30 October. This is preliminary evidence, not instructor acceptance.

Main method: Gold-YOLO head detection followed by 6DRepNet360 full-rotation estimation. It needs no facial landmarks or per-input seat calibration in automatic mode. Baseline: MediaPipe facial landmarks plus eye/nose/chin ratios. Both run on the identical prepared images.

Pipeline: decoded BGR image -> model-specific preprocessing -> head/face detection -> rotation or geometric ratios -> direction and explicit unknowns -> annotated PNG/AVI plus JSON labels and counts. Counts are observations, not unique identities. Saved video uses source-frame timestamps and requires constant frame rate.

Head preprocessing: whole-image 640-square RGB /255 tensor; detected crop expanded 1.2x, resized to 256, center-cropped to 224 and ImageNet-normalized. The reconstructed rotation maps the forward head axis into camera axes. Fixed yaw/down thresholds are 25 degrees; mixed side/down, upward, poor-quality and unsupported poses remain unknown.

Fixed set: 63 images from 15 base frames and 3 target subjects/recordings. Twelve portrait bases each have original, low-light, blur, occlusion and reduced-size variants (60 images); 3 masked classroom crops add natural clutter. No local model fitting or threshold tuning used these sources. Labels were visually assigned before inference; human cross-check is pending. Final Pointing04 test images remain locked.

These are single-subject classification examples, mostly portrait stress proxies. Adjacent frames and variants are correlated; 63 images do not mean 63 independent subjects. Labels describe visible head direction, not measured angles or eye gaze. Synthetic conditions are explicitly identified. Upstream pretraining overlap cannot be fully audited.

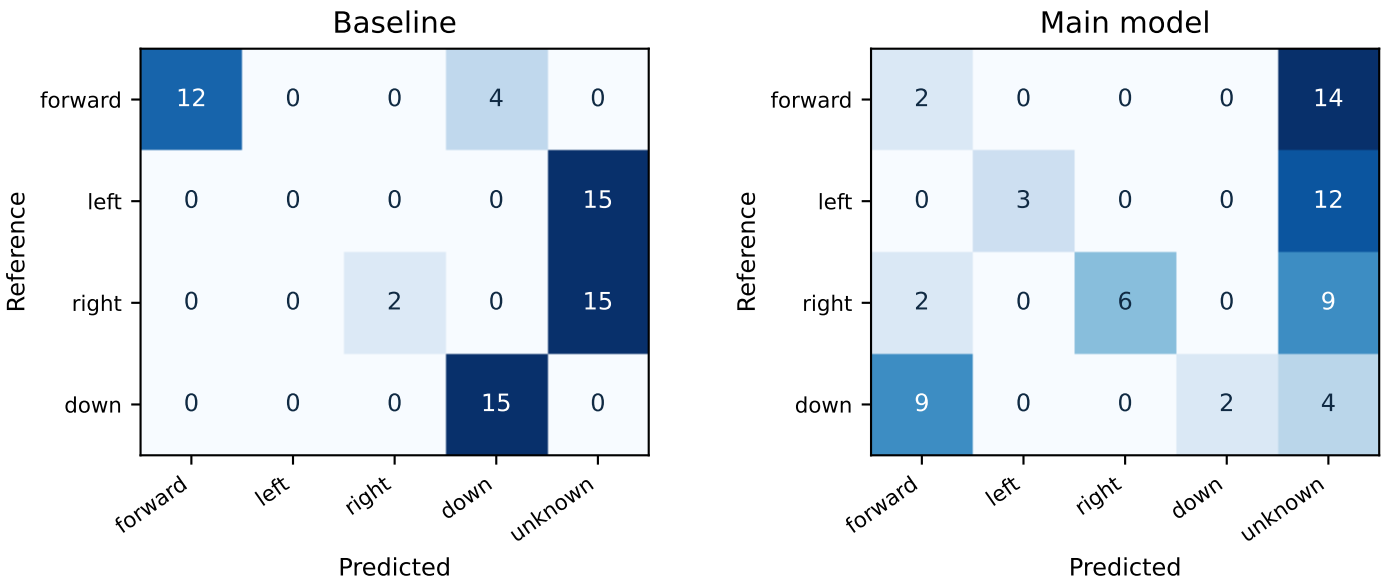
```
python scripts/run.py --input teacher-inputs --output-dir outputs/runs/live --method head
```

Measured comparison on identical inputs

Method	Correct	Accuracy	Macro-F1	Coverage	ms/image	images/s
Baseline	29/63	46.0%	0.488	52.4%	7.66	130.54
Main: head model	13/63	20.6%	0.307	38.1%	83.48	11.98

Accuracy includes all 63 references; missed detections and unknowns are incorrect. Macro-F1 averages four fixed classes, including zero-support handling. Coverage is usable directions / all references. More than one detection makes a single-subject image ambiguous; no best-face selection. These metrics are not multi-person detection AP or event accuracy.

Original images: baseline 8/15, main 8/15. On six unmodified portrait side profiles the main model gets 6/6 correct, baseline 1/6; this subgroup benefit does not establish overall improvement. Overall main accuracy is lower. No thresholds were changed after seeing these results.



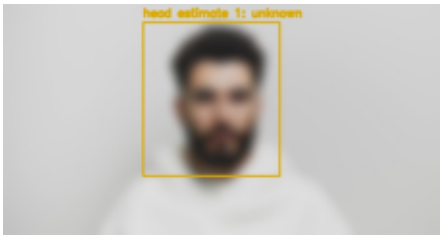
Rows are reference classes; columns are predictions. Main model: 39/63 unknown; 9 down references are predicted forward. A large unknown column indicates observation failure or rejection, not correct classification. The fixed 25-degree down threshold leaves the original downward examples (estimated about 20-22 degrees) labeled forward.

Laptop: Apple M5 MacBook Air, arm64, Python 3.12.15, macOS-27.0.1-arm64-arm-64bit. CPU inference; sequential baseline then main. Timing includes decode, preprocessing, inference and record construction; excludes model load, checksum preflight and rendering/writes. No explicit warmup excluded. This is one initial run; it is not whole-video real-time FPS.

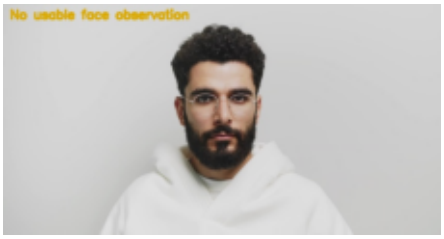
Twelve recorded main-model failures

Deterministic round-robin selection across input conditions; every panel is an actual recorded error. Labels show reference -> prediction. Gallery selection affects presentation only; aggregate counts include every failure. See gallery.json for full predictions and reasons.

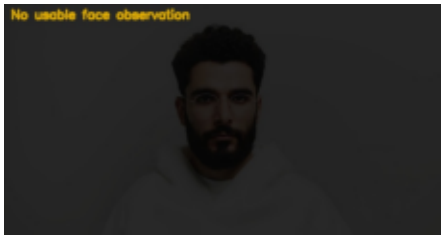
1. blur
forward -> unknown



2. clean_pose
forward -> unknown



3. low_light
forward -> unknown



4. mask_and_classroom_clutter
right -> forward



5. occlusion
forward -> unknown



6. small
forward -> unknown



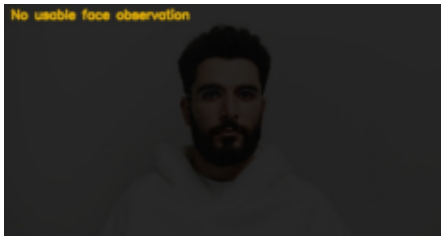
7. blur
forward -> unknown



8. clean_pose
forward -> unknown



9. low_light
forward -> unknown



10. mask_and_classroom_clutter
right -> forward



11. occlusion
forward -> unknown



12. small
forward -> unknown



Sources: cottonbro studio (Pexels 5474326), Nicola Barts (7924513), RDNE Stock project (8363848). Pexels license permits use/modification; no endorsement is implied. Images illustrate model behavior only.

Error groups, next changes and reproduction

Main-model failures by assigned input condition (50 total): blur: 12; clean_pose: 5; low_light: 12; mask_and_classroom_clutter: 2; occlusion: 10; small: 9. These are observed conditions, not proven causal diagnoses. Recorded unknown reasons distinguish detector misses, blur rejection and unsupported pose estimates.

1. Blur/low light: test a better head detector and validated preprocessing on a separate development set. Expected effect: higher detection coverage and fewer unusable poses on a new stress set; target at least +10 percentage points of coverage without lowering overall accuracy.

2. Downward and mild sideways poses: obtain independent angle/ordinal labels and select class boundaries on development data. Expected effect: fewer down -> forward and right -> forward errors. Target at least +0.10 down/right recall on a new held-out set; never select thresholds using this CP3 set.

3. Occlusion and reduced-size heads: compare multiscale detection and temporal evidence with explicit gaps on labeled development clips. Expected effect: fewer missed heads and less jitter, while keeping unknowns explicit. Target +10 points coverage with no increase in false review events; measure speed cost separately.

Reproduce: install the pinned base and head requirements, download verified models and the three pinned source videos, run prepare_cp3.py, then evaluate_cp3.py with the versioned manifest and frozen protocol. Regeneration verifies image hashes without rewriting the manifest. The submission index supplies exact commands. Numbers come from outputs/evidence/cp3-core/comparison.json; no manual table entries replace evaluator measurements.

Limits and remaining checks: three subjects, correlated frames, synthetic stress and pending human label review limit generalization. Main method does not improve overall baseline accuracy, so the excellent primary-method rubric tier is not established. Teacher-provided inputs and each member's explanation remain in-class checks. Seat/event accuracy and live camera support are outside this automatic-mode result.

Explain one failure: trace detection score/quality gate -> pose angles -> 25-degree decision -> output. Know why a masked face may defeat landmark geometry, why a head can be detected but direction rejected, and why head orientation cannot prove where the eyes are looking. Every member must rehearse their own explanation.